

1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

Because the MFCC is able to extract the features better of the audio message and scale the frequencies from the audio signal to the way the human ear interprets them. This way the model can more accurately train to recognize the sort of features that humans listen for in audio.

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

The representative data set allows the model to accurately scale the floating point values into the 8 bit range (-128 → 127) by showing what the range of the floating point values will be approximately. This way the model can most accurately and efficiently use the int8 range. The representative dataset allows the model to use a small sample of data to find a good zero value and min/max that fits the broader dataset without blindly guessing or overly generalizing.

3. What does `audio_processor.get_data()` do? Refer to `input_data.py`.

The `get_data()` function does the data loading and preprocessing for the `AudioProcessor`. It selects an audio file, applies time shifting, can add noise, computes the MFCC, and returns a numpy array of feature vectors

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are `g_model` and `g_model_len`, and why do they need to be copied correctly into the Arduino source file?

The `g_model` is the actual bits of the model in hex form. The `g_model_len` is important to tell microcontroller how many bytes to read when flashing.

5. Please include a screenshot showing the correct identification of at least one keyword, either **“YES”** or **“NO”**, as displayed in the Arduino IDE Serial Monitor after deploying your KWS model on the Arduino Nano 33 BLE. The screenshot should be included in your submitted PDF.

I was not able to get mine working I kept having an error that we were not able to solve in class.